

Advanced Packet Sniffer + ARP Spoofing Detector

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Project Summary

Designed and implemented an Advanced Packet Sniffer and ARP Spoofing Detector using Python and Scapy. The project captures live network traffic, performs protocol analysis, monitors ARP packets, detects suspicious IP-to-MAC mapping changes, and generates security logs to support network monitoring and intrusion detection activities.

Abstract

This project implements a packet sniffer and ARP spoofing detector using Python, Scapy, and Npcap.

Project Objectives

Capture packets, analyze protocols, monitor ARP traffic, detect spoofing, and generate logs.

Tools and Technologies

Python, Scapy, Npcap, VS Code, Windows.

Methodology

Live packet capture, protocol analysis, ARP monitoring, and alert generation.

Implementation

packet_sniffer.py and arp_detector.py modules were developed.

Testing and Results

Traffic capture, protocol detection, ARP monitoring, and logging were successfully tested.

Conclusion

The project demonstrates practical network monitoring and intrusion detection concepts.